

PARTICULAR SPECIFICATION P6

Instrumentation Control and Automation (ICA) Works

CONTENTS

P6.1	Introduction	1
P6.1.1	Overview	1
P6.1.2	Locations of ICA Works	1
P6.1.3	Interfaces and Interactions with other Parties	1
P6.2	Scope of Work	2
P6.2.1	ICA Scope	2
P6.2.2	ICA Equipment and Interfacing Work by Others	7
P6.3	Control System Architecture Requirements	9
P6.4	Control System Functional Requirements	9
P6.4.1	General System Requirements	9
P6.4.2	PLC & SCADA Software	10
P6.5	Installation Requirements	12
P6.5.1	Hazardous Areas	12
P6.5.2	Safety Instrumented Functions	12
P6.5.3	ICA Cabling	12
P6.5.4	Cable Support System Beneath Raised Floor	14
P6.5.5	Instrumentation Piping and Fittings	15
P6.5.6	Instrument and Actuator Installation Requirements	15
P6.5.7	Lightning and Surge Protection	16
P6.5.8	Instrument Earthing	16
P6.5.9	Contractor to Finalise Equipment Locations	17
P6.5.10	Equipment Requiring UPS Power	17
P6.6	PLC Hardware for Design and Build Vendor Packages (Digestion Pre-Treatment System and Sidestream Nitrogen Removal Facility)	18
P6.6.1	General	18
P6.6.2	PLC Processors	19
P6.6.3	PLC Communications	20
P6.6.4	Vendor Package Design and Build PLC Input / Output	21
P6.6.5	Vendor Packages Design and Build Interface with Switchboards and Motor Control Centres	22

P6.6.6	Operator Interfaces	24
P6.6.7	Engineering Laptop.....	24
P6.6.8	Interface with IWMF.....	24
P6.7	Data Communication Networks.....	25
P6.7.1	Background	25
P6.7.2	Field Networks	26
P6.7.3	Connections to PAN Subring Network.....	27
P6.8	Instrument and Control Panels.....	27
P6.8.1	General	27
P6.8.2	Outdoor ICA Panels	28
P6.8.3	RIO Panels	29
P6.9	Instrumentation	29
P6.9.2	Analyser Panels	30
P6.9.3	Instrument Power Supply	31
P6.9.4	Vendor Package Design and Build Instrumentation.....	31
P6.9.5	Independent Rooms Temperature Sensors	31
P6.10	Infrastructure For Wireless Access Point Network	32
P6.10.1	Infrastructure Locations	32
P6.10.2	Infrastructure Requirements	32
P6.11	Power Monitoring System (PMS).....	33
P6.12	Asset Management System (AMS).....	34
P6.13	Testing & Commissioning	35
P6.14	Contractors Submittals.....	36
P6.15	Appendices	37

NOTE: THIS DOCUMENT SUPERCEDES THE ORIGINAL VERSION OF THE P6 – INSTRUMENTATION CONTROL AND AUTOMATION (ICA) WORKS AT THE CLOSE OF TENDER. THE CHANGES HAS BEEN HIGHLIGHTED IN YELLOW.

P6.1 INTRODUCTION

P6.1.1 Overview

- (a) The Specification and Drawings describe the minimum requirements for the instrumentation, control and automation (ICA) elements of the Works. All instrumentation and control equipment shall comply with General Specification, G6. This Specification shall also be read in conjunction with all other particular and general specifications.
- (b) Some elements and components will be supplied by Other contractors, and for these the Contractor shall be responsible for incorporating these components into the Works.

P6.1.2 Locations of ICA Works

- (a) ICA works shall be provided in the following locations:
 - (i) Biosolids (WSB00/ WSB01); and
 - (ii) Digestion Pre-treatment 1 (WSD11);
 - (iii) Sidestream Nitrogen Removal (WST02).

P6.1.3 Interfaces and Interactions with other Parties

- (a) In addition to the general interfacing requirements specified in Particular Specifications P1, the Contractor will need to interface significantly with ICA works by the following contracts:
 - (i) Contract 5A (C5A) - Plant Monitoring and Control System;
 - (ii) Contract 1A (C1A) - Site Development Works;
 - (iii) Contract 4A (C4A) – Biosolids and Digesters;
 - (iv) Contract 6B (C6B) – Site Completion;
 - (v) Contract 6A (C6A) – Admin and O&M Building; and
 - (vi) Integrated Waste Management Facility (IWMF) Project.
- (b) The Contractor shall work with the S.O. and the above contractors to facilitate interfacing. As a minimum the Contractor shall:
 - (i) For C5A:

- 1) Grant control to C5A of XLV Rooms, UPS & UPS Battery Rooms at appropriate times for the installation of substantial amounts of equipment over an extended period including installation of sitewide backbone fibre cables; and
 - 2) Jointly work with C5A to achieve interface of Digestion Pre-Treatment, and Sidestream Nitrogen Removal Facility SCADA, PLCs process control and field control network including communications to overall TWRP control and monitoring provided by C5A;
- (ii) For C5A, C6A and C6B:
- 1) Provide cabling for subsequent use by C5A, C6A and C6B in the implementation of various systems as specified.
- (iii) For C4A and C5A:
- 1) Jointly work with C4A and C5A to provide process interface from DWAS silos feed, digesters, LPHW, IWWMF and other interfaces required for Digestion Pre-Treatment process control.
- (iv) For C1A:
- 1) Jointly work with C1A to achieve interface of ICA cable duct banks and manholes at Sidestream Nitrogen Removal Facility;
- (v) For IWWMF:
- 1) Jointly work with S.O. and NEA contractors to provide process interface from IWWMF to Biosolids Building.
- (c) The Contractor shall interface and work together with the S.O. with regard to the S.O.'s development, testing, and commissioning of PLC & SCADA software for integration with TWRP's sitewide monitoring and control system.
- (d) The Contractor shall consider space for future panels in various ICA rooms (Process XLV rooms, Non-Process XLV rooms, UPS rooms and Battery rooms) within this Contract.

P6.2 SCOPE OF WORK

P6.2.1 ICA Scope

The Contractor shall provide ICA works according to the Specifications and Drawings, including but not limited to:

- (a) The Contractor shall provide fully automated Digestion Pre-treatment and Sidestream Nitrogen Removal design and build facilities, which interface seamlessly with TWRP overall monitoring and control system.

- (b) Interface between concrete encased cable duct banks for communication cables and ICA backbone network duct banks and manholes provided by C1A, including but not be limited to:
 - (i) Banks of six (6) 110 mm diameter uPVC pipes encased in concrete (duct banks) and buried, and any associated building entry manholes, to connect from sitewide backbone cable duct banks and field cable duct banks to the buildings as shown in the Drawings, including entry to Sidestream Nitrogen Removal Facility (WST02).
- (c) Refer to Particular Specification P3 Civil and Structural Works for civil works requirements for cable duct banks and manholes.
- (d) The Contractor shall provide PLCs, PLC programming, engineering works including an updated PCN, an FDS, a peer-to-peer data exchange schedule, a SCADA data exchange schedule, liaison with S.O.'s SCADA programmers, all testing and commissioning documentation, for the following PLCs:
 - (i) PLCs controlling the Digestion Pre-treatment including unit PLCs controlling each THP Train; and
 - (ii) PLCs controlling the Sidestream Nitrogen Removal including master PLCs and unit PLCs controlling Sidestream Nitrogen Removal aeration blowers;
- (e) The Contractor shall provide on-site-HMIs, software configuration for on-site HMI's (each minimum of 15"), including graphics, database, alarms and trends for the following:
 - (i) Digestion Pre-treatment on-site HMIs (Human Machine Interfaces) including on-site HMIs for each THP Train;
 - (ii) Sidestream Nitrogen Removal on-site HMIs;
 - (iii) Sidestream Nitrogen Removal aeration blower master on-site HMIs;
- (f) The Contractor shall provide two (2) new engineering laptops, one (1) for Digestion Pre-treatment and one (1) for Sidestream Nitrogen Removal Facility. Each engineering laptop shall be provided with necessary software for relevant PLC and on-site HMIs programming, software development and configuration. The Engineering Laptops' software shall be licensed to the Board and handed over new to the Board after the completion of the Works.
- (g) The Contractor shall provide complete fully functioning compressed air system controllers with required network interface to TWRP's monitoring and control network and system.

- (h) The Contractor shall provide technical input into the S.O.'s development of control system software, including but not limited to:
 - (i) The Contractor shall attend a CHAZOP (Control System HAZOP) workshop facilitated by C5A including providing technical support for network interface from this Contract to overall TWRP control and monitoring network. Attendees shall include the process safety system designer and control technical representative from the Digestion Pre-treatment system and a process or control technical representative of the manufacturer of the Sidestream Nitrogen Removal system;
 - (ii) Plant SCADA development, programming, testing and commissioning at TWRP's SCADA will be provided by S.O.. The Contractor shall liaise and coordinate with C5A and S.O. for Digestion Pre-treatment, THP Trains and Sidestream Nitrogen Removal data mapping and full SCADA integration with TWRP plant SCADA.
- (i) The Contractor shall conduct, facilitate and participate in workshops for Digestion Pre-treatment and Sidestream Nitrogen Removal Facility Process Control Narrative (PCN) review and for Functional Design Specification (FDS) development. The workshops shall be coordinated with the S.O. to align with overall TWRP PCN and FDS standards. Attendees and contribution shall include process technical representatives from the Digestion Pre-treatment process specialist, the Sidestream Nitrogen Removal process specialist and the Sidestream Nitrogen Removal aeration blower system manufacturer.
- (j) All field devices including actuated valves and process instrumentation including analysers, analyser panels and remote transmitters and indicators. The field devices shown in the P&IDs are the minimum process requirements and functionality based on the reference design. If the Design and Build process equipment provided by the Contractor requires additional instrumentation or requires additional actuators to be automated for proper functioning of a complete Plant, then the Contractor shall provide these additional instruments, actuated valves and field devices and the cost shall be deemed to be included in the Contract Sum;
- (k) Enclosures, with appurtenances and internal wiring, for control system hardware equipment. Indicative panel locations are as shown on the Drawings, the Contractor shall design the panels and locations of panels as required by Design and Build requirements. Enclosures to be provided by this Contract include:
 - (i) Remote Input/ Output (RIO) panels. RIO panels shall be provided with all required power supplies, communication modules, input/ output (I/O) modules, terminal blocks, wiring and network hardware for the panels' interface to the Field Network; and
 - (ii) Panels complete with PLCs, I/O and on-site HMIs for the Sidestream Nitrogen Removal aeration blowers, Digestion Pre-treatment system and THP Trains, and Sidestream Nitrogen Removal system.

- (l) Local control stations for inaccessible actuated valves, inaccessible penstocks, and inaccessible slidegates;
- (m) UPS Field Distribution Boards (DBs) as required for effective and reliable UPS power distribution;
- (n) Process field cabling including:
 - (i) Power, signal and control cabling to/from field devices;
 - (ii) Power supply cabling to ECCPs, LCPs and RIO panels;
 - (iii) Copper and fibre optic (F/O) cabling for field control networks connecting Process XLV Rooms and LV Switchrooms to Digestion Pre-treatment Master PLCs panel, THP Train unit PLCs, Sidestream Nitrogen Removal master PLCs Panel, Sidestream Nitrogen Removal aeration blowers master and unit PLCs Panels, RIO panels, and MCCs panels;
- (o) Cabling connecting on-site HMIs to design and build PLCs, ECCPs and LCPs including Digestion Pre-treatment Master PLCs panel, THP Train unit PLCs, Sidestream Nitrogen Removal master PLCs Panel, Sidestream Nitrogen Removal aeration blowers master and unit PLCs Panels;
- (p) UPS Power cables to Sidestream Nitrogen Removal Master PLC panel, Sidestream Nitrogen Removal Aeration Blower Master Control Panel (MCP) and Local Control Panels (LCPs), Digestion Pre-treatment Master PLCs, THP Train unit LCPs, RIO panels, field instruments, non-process loads and other loads as specified;
- (q) Network Panels with FOBOTs for non-process cables;
- (r) Any Network Panel with FOBOTs for fire systems cables;
- (s) Cable support systems for all ICA cabling including cable support system beneath raised floor;
- (t) Fibre Optic Breakout Trays (FOBOT) for termination of all F/O cables and Fibre Optic Interface Panel racks to house the FOBOTs;
- (u) Non-Process Cabling for the following systems:
 - (i) Plant Wireless (Wi-Fi) Access Point Network including Field Fibre Optic Interface Panels (FFOTP); and
 - (ii) Cabling from Power Monitoring System (PMS) at process and non-process Electrical panels / MCCs provided by this Contract;
- (v) Sensors and instrumentation, including field cabling and cable support systems;

- (w) Edge switches for the Power Monitoring System (PMS) at electrical panels provided by this Contract as shown on the Drawings;
- (x) Any machines or equipment having integral PLCs and/or on-site HMIs shall be provided fully programmed, tested and commissioned. The Contractor shall ensure that all components and equipment of the computer system that are associated with their scope of works shall be hardened based on the System Security Standards/ Guidelines document provided by the Board/ OEM;
- (y) The Contractor shall be obliged to protect all works provided in this Contract against cybersecurity threats:
 - (i) All laptops / PC shall be installed with licensed Anti-virus software and updated with the latest definition. They shall be scanned for malware using the licensed Anti-Virus software before carrying out any works and show proof of the scan to the Board;
 - (ii) Only authorised devices are allowed to connect to the SCADA networks. No unauthorised devices are allowed to connect to the SCADA network. The Contractor shall only use notebooks that are approved by the Board. The approved notebooks that are used for interfacing with SCADA shall have an updated valid Anti-Virus software installed in it. The approved notebooks shall be scanned and be free of any viruses before entering the TWRP Plant for use at TWRP Plant site. All changes shall be made on the Board's approved notebook, SCADA maintenance / engineering workstation and/or any SCADA computer / server;
 - (iii) The Contractor shall provide a security strategy to protect and isolate the systems from being affected by viruses via removable media such as USB flash drive storage devices;
 - (iv) All USB ports shall have physical locks with procedures set in place for access to the USB ports or locked down at the OS level via a central management software and antivirus control; with procedures set in place to allow only appropriate level of authorization to perform the unlocking of the USB ports.
- (z) BIM data for permanent assets;
- (aa) Final configuration back-up of all field instruments and systems including but not limited to ID/Tag name, Communication Settings and other settings, to be submitted to the S.O.. One of the soft copies shall be in the form of files uploaded to a backup server as directed by the S.O., the remaining two soft copies shall be as DVD-ROM;
- (bb) As-built drawings as specified below in "Contractors Submittals";

- (cc) Upon substantial completion of the Works (or parts thereof) and when all relevant testing and commissioning has been completed to the satisfaction of the S.O., the Contractor shall submit the software configurations for all controllers, on-site HMIs, network devices, variable speed drives and instrumentation calibration reports in a manner which shall be accessible to the S.O. The Contractor shall upload this information to the BIM and asset management system in accordance with General Specification G1.2 Information Management and Particular Specification P2 Preliminaries and General;
- (dd) The Contractor shall handover to the S.O. two copies of application software licensed in the Board's name, including the original storage media with the original manufacturers' documentation, of any software required to modify the Digestion Pre-Treatment Master PLCs and on-site HMI's, THP Train unit PLCs software and on-site HMI's, Sidestream Nitrogen Removal master PLCs and on-site HMI's, Sidestream Nitrogen Removal aeration blower master control panel PLC software and on-site HMIs and Sidestream Nitrogen Removal aeration blower unit PLC software and on-site HMIs;
- (ee) The Contractor shall provide, within 6 (six) months of Award, one of each of the following listed items, assembled and enclosed in a manner suitable for use in a bench testing environment, and complete with flexible power cords and plugs, to enable the S.O.'s MCS programming team to complete control system integration development and testing:
 - (i) One power monitoring device;
- (ff) Two (2) sets of all Contractor-engineered controller software (program code) shall be submitted to the S.O. in DVD-ROM disks and a third set shall be uploaded to a backup server as directed by the S.O.. Contractor-engineered (including vendor-engineered) controller software shall be unlocked and without any access-controls to function blocks;
- (gg) Training of the Board's staff shall be provided in accordance with General Specifications G2 Preliminaries. This shall include but not be limited to:
 - (i) Training in ICA hardware maintenance & troubleshooting;
 - (ii) Training in use of provided software tools;
 - (iii) Training in the structure and organisation of contractor-engineered controller software; and
 - (iv) Training shall be provided for two (2) sessions after testing and commissioning or at a period/timing as instructed by the S.O.

P6.2.2 ICA Equipment and Interfacing Work by Others

The following work will be provided under C5A:

- (a) Supply and configuration of SCADA and Network Hardware as follows and as further detailed elsewhere. The elements and components supplied by others do not include panel enclosures, terminals, wiring, panel power supply units and other panel appurtenances, unless stated otherwise in (b) below:
 - (i) Process Ethernet Backbone Network switches;
 - (ii) Non-Process Ethernet Backbone Network switches;
 - (iii) Edge Ethernet switches and wireless access point units for Wi-Fi access point network;
 - (iv) SCADA I/O servers and clients; and
 - (v) Sub-ring ethernet switches in PAN Networks.
- (b) Supply and installation of the following panels as shown in Drawings:
 - (i) Server racks (SVRR) in Process XLV Room;
 - (ii) Fibre Optic Interface (Patch) Panels and FOBOTs in Process and Non-Process XLV Rooms for process and non-process backbone and fire systems backbone; and
 - (iii) Selected FOIP containing PAN subring switches and FOBOT's for PMN/PAN fibre optic backbone cabling as specifically identified in the Drawings.
- (c) Supply and installation of fibre optic backbone cabling, including FOBOT's, as shown in the Drawings:
 - (i) Process Automation Network and Process Monitoring Network sitewide backbone;
 - (ii) Non-Process sitewide backbone; and
 - (iii) Fire System backbone.
- (d) Supply and installation of UPS Systems with external Bypass Switch and including cabling to UPS unit and Bypass switch and the supply and installation of UPS Main Distribution Board(s) for the following facilities:
 - (i) Biosolids Building; and
 - (ii) Sidestream Nitrogen Removal Building.
- (e) Network Monitoring System including for Field Networks.

- (f) The following work will be provided under C5A:
 - (i) IT hardware for Plant Wireless (Wi-Fi) network; and
 - (ii) PMS main servers hardware and user interface platform.

P6.3 CONTROL SYSTEM ARCHITECTURE REQUIREMENTS

- (a) The Contractor shall provide control and monitoring networks for Digestion Pre-Treatment System and for Sidestream Nitrogen Removal, to achieve a fully functioning control system as per the requirements for design and build facilities/plants as specified in G6.17 and as described below in P6.6 and as shown on Drawings.

P6.4 CONTROL SYSTEM FUNCTIONAL REQUIREMENTS

P6.4.1 General System Requirements

- (a) Sidestream Nitrogen Removal Aeration Blowers are considered complex machines (refer General Specification G6.17 PLC Hardware for Machine PLC Specification).
- (b) Digestion Pre-treatment System master PLCs, THP Train unit PLCs and Sidestream Nitrogen Removal master PLCs shall be considered to be vendor package design and build facilities PLCs (as per G6.17) and shall comply with section P6.6.
- (c) Digestion Pre-Treatment master PLCs, THP Train unit PLCs and Sidestream Nitrogen Removal master PLCs and Sidestream Nitrogen Removal aeration blowers PLCs shall provide automatic control as described in the following Process Control Narratives (PCNs):

Appendix	PCN Name
P6A	Digestion Pre-treatment and Sidestream Nitrogen Removal PCN
P6B	Standard Control Features PCN

- (d) The Contractor shall refer to 'Standard Control Features PCN' control narrative in Appendix P6B for the Hierarchy of Manual Operation.
- (e) The Contractor's design and build control and monitoring equipment shall be compatible with C5A's network monitoring system provided by ABB;
- (f) Compressed Air Systems including any proprietary master controllers by the compressor manufacturer, are not considered to be PLC's but shall still have an Ethernet interface to the PAN. Refer to Particular Specifications P4 Mechanical for compressed air systems requirements.

P6.4.2 PLC & SCADA Software

- (a) Digestion Pre-Treatment system and Sidestream Nitrogen Removal Facility system shall be provided as fully automated facilities with required interfaces to TWRP's overall monitoring and control system.
- (b) The Contractor shall carry out the PLC programming of Digestion Pre-Treatment plant PLCs and Sidestream Nitrogen Removal System PLC's.
- (c) The Contractor shall configure Digestion Pre-Treatment HMI graphics at Digestion Pre-Treatment Master on-site HMI and each THP Train unit on-site HMI(s), Sidestream Nitrogen Removal aeration blowers on-site HMIs and Sidestream Nitrogen Removal master on-site HMI(s).
- (d) The Contractor shall coordinate and work together with the S.O. and C5A for interfacing of Digestion Pre-Treatment Master PLCs, THP Train unit PLCs, Sidestream Nitrogen Removal aeration blowers (master and unit) PLCs and Sidestream Nitrogen Removal master PLCs and all required on-site HMI(s) SCADA graphics as directed and approved by S.O. to integrate seamlessly with TWRP SCADA overall monitoring and control system.
- (e) The Contractor shall have obligations related to the development of the process control narratives (PCNs) and shall provide the functional design specification (FDS) and obligations related to the software and hardware FAT, SAT, and commissioning for the Digestion Pre-treatment plant and Sidestream Nitrogen Removal Facility as follows. The Contractor shall:
 - (i) Review and update the PCNs provided after award, to make the PCN reflect the Contractor's design and build proposed process equipment. Draft versions of the PCNs are provided in Appendices P6A and P6B;
 - (ii) The Contractor shall provide FDS's for the complete Digestion Pre-treatment system; Sidestream Nitrogen Removal system; and aeration blowers for sidestream nitrogen removal and compressed air system. These FDS's shall follow the format of the S.O.'s FDS's for other areas of the plant;
 - (iii) Conduct control system software and hardware FAT and SAT according to design and build Contract requirements including full technical support from Contractor's ICA Engineer and technical representatives from Digestion Pre-treatment system and Sidestream Nitrogen Removal manufacturers;
 - (iv) Attend and provide technical support for S.O.'s SCADA software FAT in Singapore to witness and provide technical support that the control system will be suitable for the Contractor's equipment and achieve design and build requirements. Attendance and technical support shall include the Contractor's ICA Engineer and technical representatives from Digestion Pre-treatment system and Sidestream Nitrogen Removal manufacturers; and

- (v) Attend and provide technical support for the S.O.'s SCADA software SAT on site to witness and provide technical support that the implemented control system is suitable for the Contractor's equipment and achieve design and build requirements. Attendance and technical support shall include the Contractor's ICA Engineer and technical representatives from Digestion Pre-treatment system, Sidestream Nitrogen Removal and Sidestream Nitrogen Removal Aeration Blowers manufacturers.
- (f) The Contractor's Initial Operation Period staff shall attend the operator training sessions conducted by S.O..
- (g) The Contractor shall fully programme and test PLCs and on-site HMI(s) for each of the Digestion Pre-Treatment master PLCs, each THP unit PLCs, Sidestream Nitrogen Removal master PLCs, Sidestream Nitrogen Removal aeration blowers Master Control Panel (MCP) and each blower Local Control Panels (LCPs). PLC and on-site HMI(s) programming shall comply with the TWRP Software Standards Specification which will be issued after Award. The required FDS format will be issued after Award.
- (h) The Contractor shall provide all Digestion Pre-treatment PLC & HMI software and hardware FAT and SAT and Sidestream Nitrogen Removal PLC & HMI software and hardware FAT and SAT at the Contractor's designated testing facility, both attended and approved by the S.O..
- (i) The Contractor shall provide fully tested Digestion Pre-treatment master PLCs and THP Train PLCs & HMI software and Sidestream Nitrogen Removal master PLCs, Sidestream Nitrogen Removal aeration blowers PLCs & HMI software for software FAT and integration to TWRP SCADA in the S.O.'s office (programming area), both attended and approved by the S.O..
- (j) The Contractor shall provide PLC & HMI programmer(s) from Digestion Pre-treatment system and Sidestream Nitrogen Removal system manufacturers as required throughout plant commissioning.
- (k) The Contractor shall maintain an electronic copy of Digestion Pre-treatment master PLCs, THP Train PLCs and Sidestream Nitrogen Removal master PLCs, Sidestream Nitrogen Removal aeration blowers PLCs programming and on-site HMIs configuration work in progress, in native file format, updated weekly, in a location readily accessible by the S.O. From the commencement of software FAT onwards, the Contractor shall maintain PLC programming native files in an automated revision management system accessible by the S.O..
- (l) The Contractor shall provide any software licenses and tools, including PLC and on-site HMIs programming software and any patching works required for their own work.
- (m) The SCADA and PLC platform used in TWRP is ABB, which are provided under a separate contract, Contract 5A. The PLC provided in this Contract shall be able to interfaced, worked and communicated seamlessly with the TWRP PLC and SCADA.

P6.5 INSTALLATION REQUIREMENTS**P6.5.1 Hazardous Areas**

- (a) Hazardous Areas are specified in Hazardous Area Classification (HAC) document included in Particular Specification, P5. The Contractor shall review and verify the HAC in relation to any adjustments which may be needed due to the Design and Build requirements and process equipment selected by the Contractor.
- (b) The Contractor shall provide a register of hazardous area electrical apparatus within the operations and maintenance data submitted under this Contract.
- (c) Drawings and documentation for hazardous area installations shall be prepared by an engineer with documented formal training and competency in IEC-compliant hazardous area designs.

P6.5.2 Safety Instrumented Functions

- (a) The Contractor shall provide SIF that are required to address the personnel safety risks presented by the selected mechanical equipment.
- (b) The Contractor shall submit a Functional Safety Report to the S.O., covering all SIF systems including any additional SIF requirements of the particular process equipment selected by the Contractor.
- (c) SIF systems shall be implemented in accordance with IEC 61508, IEC 61511, IEC 62061.
- (d) The Contractor shall provide all hardware and software required to implement all SIFs. This shall include PLC's if the Contractor chooses to use safety-PLC's as logic solver's for SIF. It is anticipated that the logic solvers will be safety relays rather than safety-PLC's.
- (e) The status of each SIF input and output shall be shown on SCADA. The interface between safety instrumented functions (SIF) and SCADA shall be either:
 - (i) Via hardwired I/O to design and build PLC; or
 - (ii) Via connection to PAN using a dedicated firewall to create a separate cybersecurity zone of protection for the safety system.

P6.5.3 ICA Cabling

- (a) The Contractor shall provide all ICA cabling complete with support and protection, necessary to interconnect instrumentation, control and monitoring circuits with the plant and equipment included in the Works to provide the specified instrumentation, control and automation facilities.

- (b) The cabling shall interconnect the entire facility including but not limited to any combination of the following:
 - (i) Valves/ gates/ penstocks actuators;
 - (ii) Pumps;
 - (iii) Switchgear;
 - (iv) Motor Control Centres (MCCs)/ Variable Frequency Drives (VFDs);
 - (v) PLCs' panels;
 - (vi) On-site HMIs;
 - (vii) Local Control Panels (LCPs)/ Local Control Stations (LCSs);
 - (viii) Remote Input/ Output (RIO) Panels, Junction Boxes;
 - (ix) Measuring instruments and sensors;
 - (x) All other equipment for which interconnecting cablings are required to achieve specified degree of automation, control and monitoring;
 - (xi) Field control networks; and
 - (xii) Power supply cabling from UPS Main DB(s) to UPS Field DB(s) and to other panels supplied under this Contract;
- (c) ICA cabling, cable support and protection shall be as specified in the Particular Specifications P5 Electrical Works, General Specifications G5 Electrical Works and G6 Instrumentation Controls and Automation (ICA) and shall meet the following additional requirements:
 - (i) Where ICA cables are buried outdoor, they shall run in uPVC Class B ducts of 110mm diameter, unless otherwise specified as per Drawings; and
 - (ii) Following cables outer sheath shall be provided:
 - 1) CAT6 – Grey;
 - 2) Modbus – Blue;
 - 3) Fibre Optic (Single Mode) – Black with Yellow Strips; and
 - 4) Fibre Optic (Multimode) – Black with Aqua Strips
 - (iii) Outer sheath for all other cable types shall be subjected to the approval of the S.O..

- (d) All ICA service space for cable routing, tiers, cable containment and ductbanks size as shown in the BIM Model are indicative only. The Contractor shall size and provide ICA cable routing, raceways and support system for all equipment as required. The Contractor shall study the general arrangement of the propose equipment and its cable termination box/ rack/ panel location, the routing and the drop off point accordingly to suit the service space as indicated in the BIM Model. The Contractor shall consider the bending radius of the cables and all adjustments of the cable route and the equipment layout. Any low-laying containments shall be equipped with covers protection and steps for crossing over and all cost for a complete installation shall be deemed to be included in the Contract Sum.

P6.5.4 Cable Support System Beneath Raised Floor

- (a) The Contractor shall provide cable support system beneath raised floor including but not limited to:
- (i) 200mm (W) x 100mm (D) PVC cable tray for electrical cables and 200mm (W) x 100mm (D) PVC cable tray for data network cables laid side by side, but at differing elevation (to allow cross overs), in a grid pattern beneath the raised floor in the following rooms:

Area:	Room provided with raised floor:
WST02 – Sidestream Nitrogen Removal	Process XLV Room
	Non-process XLV room
	UPS room
	Battery room

- (b) Grid intersections shall use proprietary intersection fittings.
- (c) Cable trays with adequate support shall be secured to the concrete floor.
- (d) For short cable runs between cable tray and penetrations in the raised floor, flexible protective conduit shall be used.

P6.5.5 Instrumentation Piping and Fittings

- (a) The Contractor shall provide instrument piping and tubing as required for the correct operation of the Works as specified in G6 Instrumentation Controls and Automation (ICA).
- (b) The Contractor shall provide diaphragm seals for pressure instruments whenever appropriate for the process conditions. The Drawings show the minimum requirement for diaphragm seals.
- (c) The Contractor shall provide pulsation dampeners for pressure instruments whenever appropriate for the process conditions. The Drawings show the minimum requirement for pulsation dampeners.

P6.5.6 Instrument and Actuator Installation Requirements

- (a) All instruments and actuators located 2m and above from the finished floor level shall be provided with a proper safe and permanent access platform for maintenance or troubleshooting. Platform design shall be as per S.O.'s approval.
- (b) All electric and pneumatic actuators located more than 1.5m above operating floor level or finished floor level shall be provided with a remote mounted Local Control Station (LCS) in an accessible location not more than 1.5m above operating floor level or not more than 1.5m above finished floor level.
- (c) Refer to Particular Specification P4 Mechanical Works and General Specifications G4 Mechanical for additional actuator installation requirements.
- (d) Pressure transmitters and pressure gauges (indicators) for chemical service and for sludge or other particular service, shall be provided with a diaphragm seal and shall be completed with isolation and drain/vent valve.
- (e) All pressure measurement instruments shall be provided with isolation valve and drain/vent valve.
- (f) All direct pressure measurement instruments shall be mounted at 3 or 9 o'clock positions on the cross sectional plane of the pipeline.
- (g) For plastic (including FRP) pipelines, pressure instrument mounting arrangements shall relieve any welded joints on the process line for additional stresses due to pipe vibration. Remote mounting with tubing connection shall be provided for this application.
- (h) Instruments shall also be installed in accordance with General Specification G6.2 Installation Requirements.
- (i) The Contractor shall provide instrument stands and brackets. Instruments shall not be mounted on handrails.
- (j) The Contractor shall provide a sun hood for each instrument mounted outdoors.

- (k) The Contractor shall provide a sun hood for each actuator which has an LCD display.

P6.5.7 Lightning and Surge Protection

- (a) The requirements for primary lightning protection for buildings and structures are contained within General Specifications G5 Electrical and Particular Specification P5 Electrical Works.
- (b) Secondary protection shall be provided for ICA equipment as follows:
 - (i) Lightning protection systems shall be provided to protect all solid state equipment and instruments from damage caused by lightning including transient over-voltage conditions caused by inductive and capacitive coupling along the communications, signal and power supply lines. These protection devices shall be additional to any devices that are integral to the individual items of equipment. Surge protection devices shall be provided at both ends of the cable and where instruments and analysers are located outdoors. Lightning protection system design shall be subject to the approval of the S.O.;
 - (ii) Instruments and analysers which are located outdoors shall be housed in properly earthed enclosures. Surge protection devices shall be provided for all power, communication and signal cables, except for Fibre Optic cables. Surge protection devices shall be installed at both ends of lines, and as close to the device being protected as possible and in the same enclosure; and
 - (iii) Surge protection devices shall be provided for all external antenna feeders.

P6.5.8 Instrument Earthing

- (a) The Contractor shall provide a separate insulated cable to connect the instrument earth (I.E.) bar in each ICA panel to an instrument earth bar at each XLV Room. Daisy chain connections of the instrument earth to ICA panels are not acceptable.
- (b) ICA panels located in LV Switchrooms shall be connected to an instrument earth bar in the LV Switchrooms.
- (c) Instrument earth bars shall be connected together by insulated earth cables.
- (d) Refer to Drawings for typical earthing schematics.
- (e) Instrument cable screens shall be connected to instrument earth only at the I/O panel end.
- (f) Separate instrument earth electrodes shall be provided.

P6.5.9 Contractor to Finalise Equipment Locations

- (a) Geographical locations of panels and equipment shown the Drawings are indicative only. The Contractor shall develop final locations whilst adhering to the functional intent of the Drawings and design and build requirements.
- (b) RIO Panels shall not be located in the following areas:
 - (i) Hazardous Areas; and
 - (ii) Areas which will become Hazardous Areas during maintenance operations involving opening of hatches and covers.

P6.5.10 Equipment Requiring UPS Power

- (a) General Specifications G6.1 requires all ICA equipment to be supplied from UPS. This shall include, but not be limited to, the following categories of equipment:
 - (i) SVRR Racks (servers, workstations and network switches);
 - (ii) PLC Panels and on-site HMIs;
 - (iii) RIO Panels (which in turn supply 24 Vdc instruments);
 - (iv) 230 Vac powered instruments and analysers;
 - (v) FFOTP Panels;
 - (vi) ECCP Panels;
 - (vii) FOIP Panels (unless all passive components);
 - (viii) MCP's;
 - (ix) LCP's;
 - (x) PMS equipment including edge switches for electrical panels serving process and non-process equipment; and
 - (xi) Some Building Services equipment as specified in Particular Specifications P7 and General Specifications G7.
- (b) The Contractor shall provide preliminary UPS load sizing and heat load (to adequately size the HVAC) for equipment in this Contract to the S.O. within three (3) months of Contract Award and final UPS load sizing by six (6) months of Contract Award.

P6.6 PLC HARDWARE FOR DESIGN AND BUILD VENDOR PACKAGES (DIGESTION PRE-TREATMENT SYSTEM AND SIDESTREAM NITROGEN REMOVAL FACILITY)**P6.6.1 General**

- (a) The Digestion Pre-treatment System and Sidestream Nitrogen Removal Facility shall be considered as vendor package forming part of a design and build facility, not as complex machines. Therefore PLCs for the Digestion Pre-treatment System and Sidestream Nitrogen Removal Facility shall be considered to be equivalent to Main Plant PLCs.
- (b) The Digestion Pre-treatment system and Sidestream Nitrogen Removal Facility PLC control systems, complete with I/O and on-site HMI, shall be provided by this Contractor.
- (c) On-site HMIs for Digestion Pre-treatment system and Sidestream Nitrogen Removal Facility operation shall be configured by the Contractor.
- (d) Vendor packages design and build Digestion Pre-treatment Master PLCs, THP Train PLCs, Sidestream Nitrogen Removal master PLCs and complex machine Sidestream Nitrogen Removal aeration blower master and unit PLCs shall be one of the following makes, or equivalent:
 - (i) Rockwell;
 - (ii) Schneider;
 - (iii) Siemens; or
 - (iv) ABB
- (e) Vendor packages design and build and complex machine PLC (including Digestion Pre-treatment Master PLCs, Sidestream Nitrogen Removal Master PLCs and complex machine Sidestream Nitrogen Removal master and unit aeration blower PLCs) peer-to-peer communication protocol to Plant PLC and vendor packages design and build and complex machine PLC communication protocol to Plant SCADA shall be Modbus TCP otherwise protocol converter is required. If the communication link to Plant PLC fails, direct control and monitoring via SCADA to the vendor packages design and build Digestion Pre-treatment Master PLCs, Sidestream Nitrogen Removal Master PLCs and complex machine Sidestream Nitrogen Removal master and unit aeration blower PLCs shall be made possible.
- (f) PLCs and any other controllers (including compressed air system controllers) provided by this Contract shall comply with Cybersecurity Act, Cyber Security Agency (CSA) requirements, the ISA / IEC Standard 62443 (ISA99) and PUB's Water Sector CII Policy for Cybersecurity Requirements ("Cyber Security Policy") for hardening requirements. PUB's Water Sector CII Policy for Cybersecurity Requirements ("Cyber Security Policy") will be provided after Contract Award.

- (g) Automatic control of the vendor packages design and build facility (including complex machine PLCs) shall be achieved as minimally described in the Process Control Narratives (PCNs) in clause P6.4.1(c).
- (h) The Digestion Pre-Treatment master PLCs, THP Train unit PLCs, and Sidestream Nitrogen Removal master PLCs and Sidestream Nitrogen Removal master and unit aeration blower PLCs shall control the vendor package equipment via discrete input and outputs and via Ethernet interface to MCC's and VFD's.
- (i) The PLC manufacturer shall have experienced, competent and certified engineers and technicians based in Singapore to provide on-site technical support on the PLCs supplied under this Contract. Under an emergency call-out, the PLC manufacturer shall have a suitable engineer or technician on-site within 2 hours of the request.
- (j) PLCs shall be chassis (backplane)-mounted plug-in modules, with separate modules for processors with memory, redundant power supply modules, input/output interfaces, and communications modules. Communications ports which are integral to processor modules may also be used, provided that there is also one unused local port for programming terminal connection.
- (k) Where a PLC is indicated in the Drawings as being hot-standby, its processors, power supplies and communications modules shall be provided in dual redundant hot-standby arrangements and it shall achieve bumpless transfer in the event of the failure of any duty unit.
- (l) Inputs for discrete signals (digital and analogue) shall be optically isolated from each other and from the PLC backplane.
- (m) All PLC hardware shall have conformally coated circuit boards. The conformal coatings shall be applied within the PLC hardware manufacturer's supply chain and shall be approximately 25-50 μm thick polymeric films which cover and encapsulate the printed circuit assemblies.
- (n) All PLC hardware shall be capable of operating in a harsh and corrosive industrial environment without requirement for air-conditioning or additional power supply filtering.
- (o) PLCs shall be provided with diagnostic capabilities to identify failures down to the module level. Diagnostic indication shall include but not be limited to notification of processor failure, input or output module failure, communications failure, and battery low.
- (p) PLCs and I/Os shall have embedded end-to-end cybersecurity including encryption unless the PLC's and I/Os are fully isolated from all other networks.

P6.6.2 PLC Processors

- (a) Where a PLC is indicated in the Drawings as being hot-standby, its processors shall be provided in a redundant arrangement. Redundant controllers may be either in separate chassis or in a combined chassis.

- (b) Failure of primary processor unit shall cause control of plant and communications with SCADA servers to automatically be transferred to backup processor within 1 second.
- (c) Where redundant controllers are required, redundant communications modules and redundant power supply modules shall be provided for each CPU chassis.
- (d) Program and data memory shall be provided with a minimum of six months integral battery or capacitor back-up. It shall be possible to replace any backup battery without memory loss.
- (e) A minimum of 40% spare programme memory and 40% spare data register memory shall be available when final commissioning is complete.
- (f) The PLC program shall either be stored in non-volatile memory or the CPU shall have additionally have EEPROM/flash memory card for automatic retrieval after power-cycling.
- (g) Where hot standby processors are used, processors shall be individually hot swappable.
- (h) PLCs shall be programmable using languages and formats that comply with IEC61131-3.
- (i) The PLC platform shall provide on-line editing during run-time without requiring re-compilation for most types of changes.

P6.6.3 PLC Communications

- (a) PLCs shall be connected to the process automation network (PAN). Refer also to G6.20 – Data Communication Networks.
- (b) It shall be possible to program any PLC from any Engineering Laptops as well as from the XLV room housing the respective PLC.
- (c) Each of the PLC's duty and standby processors shall have two communication links to the dual ring plant automation network (PAN), as shown on the Drawings. Thus there will be four network connections between a hot-standby PLC and the PAN.
- (d) Communication failure at one network node shall not affect communications at any other node. The PLC operating system, or standard function blocks provided, shall natively manage changeover of peer-to-peer communication message traffic between the redundant process automation networks.
- (e) PLCs shall contain sufficient communication ports such that a portable programming device can be connected directly to the CPU and operated locally without disconnecting the PLC from any permanent network.
- (f) Comprehensive communication failure alarms shall be configured.

P6.6.4 Vendor Package Design and Build PLC Input / Output

- (a) The Contractor shall design all I/O systems necessary for the correct, efficient and safe operation of the process equipment.
- (b) I/Os shall be interfaced to the field digital and analogue signals from instruments and equipment including electrical switchgear.
- (c) No failure of any single component or module shall cause any loss of function except that failure of an I/O module may cause loss of functions associated with that I/O module.
- (d) Conventional I/O with HART shall be used.
- (e) PLCs shall be fitted with input and output modules to interface with field devices. Input and output circuits shall be arranged on plug-in modules to facilitate replacement. Plug-in input and output modules shall be fitted with connectors which allow replacement of the module without disconnection of interface wiring.
- (f) Input and output modules shall be hot-swappable. i.e. They shall be able to be changed without affecting adjacent cards and without switching off the associated rack/back-plane.
- (g) All input channels, both analogue and digital, shall be isolated from one another so as to not be affected by differing common mode potentials. Analogue inputs shall be wired and configured in differential mode.
- (h) Inputs and outputs for any set of duty/standby equipment shall be arranged over multiple I/O cards such that even after the failure of a single I/O card, operation and monitoring by the PLC of at least 50% of the units shall still be possible.
- (i) When commissioning is complete, PLCs shall have a minimum, at each I/O panel, of 20% installed spare digital inputs and outputs, and 30% installed spare analogue inputs and outputs, all wired down to terminals. Input and output capacity shall be further expandable in that there shall be sufficient spare slots to allow the installation of an additional 30% of I/O cards. The spare slot requirement shall be subject to a minimum of 4 spare slots per panel.
- (j) The Contractor shall be responsible for considering each output load in detail to ensure that the electrical current capabilities of the PLC output channel are adequate.
- (k) Analogue input and analogue output modules shall be 4-20 mA HART and HART data shall be passed through the PLC to SCADA.
- (l) Instruments and valve actuators shall be connected to the PLC using 4-20 mA HART and 24 Vdc discrete signals.

(m) I/O Allocation

(i) I/O shall be allocated in a manner such that the failure of any single I/O module cannot disable more than one unit of plant. Inputs and outputs for any set of duty/standby equipment shall be arranged over multiple I/O cards such that even after the failure of a single I/O card, operation and monitoring by the PLC of at least 50% of the units shall still be possible. For example, such a failure shall not disable more than:

- 1) One process unit/ train (e.g. one screen) or;
- 2) One blower (per system) or;
- 3) One pump (per pump station) or;
- 4) One VFD or;
- 5) One chemical dosing unit, etc.

(ii) No single failure shall prevent the automatic operation of both the duty and standby unit for any equipment.

(n) P&ID Drawings for local control station types define the required interface signals for valve actuators.

(o) The Contractor shall include any I/Os as required for the proper functioning of the complete system and to satisfy the Contractor's Design and Build obligations.

P6.6.5 Vendor Packages Design and Build Interface with Switchboards and Motor Control Centres

(a) Interface for Motor Control

(i) P&ID Drawings for local control station types define the required interface signals for motors.

(ii) Interface between PLCs and motor starters, including variable speed drives, shall be by Ethernet based field network as shown in the Drawings, using one of the following protocols for monitoring and control subject to the approval by the S.O.:

- 1) Profinet; or
- 2) MODBUS TCP.

- (iii) The Contractor shall provide control data interfaces at intelligent motor control centres to achieve the Design and Build control system architecture requirements provided by this Contract and notionally as shown in the Drawings. Control data interfaces for intelligent motor control centres shall include, but not be limited to the following:
 - 1) Ability to communicate natively, without the use of any third-party protocol converter;
 - 2) Ability of MCC equipment to “ride-through” temporary losses of communications such as may occur during changeover from duty to standby PLC CPU, or remapping of traffic in ring-networks following a network media failure;
 - 3) Ability to exchange control commands, status feedback, and to download drive parameters and configurations; and
 - 4) Capacity of the control system architecture to process and convey the volume, rate, and quantity of data generated by the MCC to the design and build control system and to the TWRP overall control and monitoring system.
- (iv) The MCC internal field control network shall be arranged so that the withdrawal of one of more motor starters only interrupts communications to the withdrawn units.
- (v) Motor starters shall be fully configured, including communications interfaces, by this Contract.
- (b) Interface for Electrical Switchgear
 - (i) Any digital input and output signals between PLCs and switchboards (other than for motor starters) shall be 24 Vdc signals. Each digital output from a PLC shall provide 24 Vdc to energise the coil of an interposing relay located in the respective switchboard cubicle. Digital inputs to PLCs shall be powered from the PLC end and switched by a potential free contact in the switchboard cubicle. Analogue signal loops shall be arranged to suit the relevant equipment and as such may be powered from either end.
- (c) Any field signals which are hardwired to MCC's, shall also be individually connected to a PLC digital input channel, unless the status of each field signal is individually available on SCADA via the MCC Ethernet connection. This requirement is so as to allow specific indication on SCADA of the status of an interlock.

P6.6.6 Operator Interfaces

- (a) The plant SCADA system will be provided by C5A and configured by the S.O..
- (b) A fully-configured on-site HMI (touch screen) for the Digestion Pre-treatment and THP Train system, Sidestream Nitrogen Removal and Sidestream Nitrogen Removal aeration blowers shall be provided by this Contract.
- (c) All on-site HMIs' communication protocol to Plant SCADA shall be Modbus TCP otherwise protocol converter is required.
- (d) On-site HMIs provided by this Contract shall comply with Cybersecurity Act, Cyber Security Agency (CSA) requirements, the ISA / IEC Standard 62443 (ISA99) and PUB's Water Sector CII Policy for Cybersecurity Requirements ("Cyber Security Policy") for hardening requirements. PUB's Water Sector CII Policy for Cybersecurity Requirements ("Cyber Security Policy") will be provided after Contract Award.

P6.6.7 Engineering Laptop

- (a) Engineering laptop shall be business-grade and shall be configured to suit the installed software but as a minimum shall have the latest Intel Core 11th Gen i7-11800H processor, 32GB RAM and 1x32GB DDR4 3200 MHz, 2TB Solid State Drive, 15" FHD 1920X1080 touch display, high performance video graphics card with minimum of 4 GB RAM, 16-bit sound card, internal speakers, Wireless Intel AX201 WLAN driver and wireless card with Bluetooth 5.2, Windows 11 Pro operating system, Microsoft Office Home & Business 2021 with Adobe Acrobat Standard 2020, Security software with 36-month subscription, VMWare Endpoint Security, keyboard, mouse and ancillaries.

P6.6.8 Interface with IWMF

- (a) C4B Digestion Pre-Treatment Master PLCs shall communicate with C4A's Interface PLCs with IWMF (WSB01-PLC-1700A/B) for signals interface required with IWMF as listed below but not limited to:

Fluid/ Service Name:	Fluid/ Service Code:	Description:
Steam High Pressure	STHP	TWRP Steam Route Available
Steam High Pressure	STHP	TWRP Steam Controlled Stop Request

Fluid/ Service Name:	Fluid/ Service Code:	Description:
Steam High Pressure	STHP	TWRP Steam Emergency Shutdown
Steam High Pressure	STHP	IWMF Steam Available
Steam High Pressure	STHP	IWMF Steam Pressure
Steam High Pressure	STHP	IWMF Steam Temperature
Steam High Pressure	STHP	IWMF Steam Mass Flow

P6.7 DATA COMMUNICATION NETWORKS

P6.7.1 Background

This subsection provides general background into the overall system architecture, including elements provided by C5A:

- (a) The monitoring and control system in TWRP is connected via sitewide communication networks in ring configurations. SCADA-PLC traffic is carried by Process Monitoring Network (PMN) and peer-to-peer PLC traffic is carried by Process Automation Network (PAN). The PAN extends to multiple locations within each facility, in the form of a sub-ring network, to connect all plant PLCs within the facility;
- (b) Sitewide PMN and PAN networks, including fibre optic cabling between buildings, and network switches, are provided by C5A. PAN sub-ring cabling will be provided by others as shown in the Drawings. PAN subring cables will be 24 core, and of the type shown in the Drawings;
- (c) The Contractor shall provide connection from Digestion Pre-treatment master PLCs, Sidestream Nitrogen Removal Master PLCs, Sidestream Nitrogen Removal Master Blowers PLCs and Sidestream Nitrogen Removal aeration blowers unit PLCs to the PAN sub-ring networks as shown in the Drawings;

- (d) PMN and PAN sitewide backbone cables generally terminate at each facility's Process XLV Room but may terminate in other rooms as shown in the Drawings. Where sitewide backbone cables terminate in rooms other than the Process XLV room, then backbone extension cabling from the C5A backbone cable termination point to the facility Process XLV room shall be provided by the Contractor. These cables shall comply with ITU-T G.652D and shall have all dielectric armour for rodent and termite protection as well as complying with the requirements of General Specifications G6.2 Installation Requirements. The proposed cable make and type shall be submitted to the S.O. for approval to ensure compatibility with the backbone cables;
- (e) Non-process backbone cables generally terminate at each facility's Non-Process XLV Room but may terminate in other rooms. These cables provide communications for building services systems and other non-process systems including power monitoring and condition monitoring systems. Where sitewide backbone cables terminate in rooms other than the Non-Process XLV room, then backbone extension cabling from the C5A backbone cable termination point to the facility Non-Process XLV room shall be provided by the Contractor. These cables shall comply with ITU-T G.652D and shall have all dielectric armour for rodent and termite protection as well as complying with the requirements of General Specifications G6.2 Installation Requirements. The proposed cable make and type shall be submitted to the S.O. for approval to ensure compatibility with the backbone cables;
- (f) The Contractor shall provide field networks interfaced at FOBOTs in Process XLV Room as shown in the Drawings, which shall extend to RIO panels and MCCs throughout each facility including outdoors and in this Contract; and
- (g) The Contractor shall set IP addresses for any equipment provided by this Contract. Addresses shall be set to those advised by the S.O..

P6.7.2 Field Networks

- (a) The field control networks shown on the Drawings is notional and indicative only. The field control network as a minimum, shall connect RIO panels and motor starters (including standalone VFD's) to Programmable Logic Controller (PLC) panels provided in this Contract in the Process XLV Room, and shall have the following properties:
 - (i) Be a fault-tolerant TCP/IP Ethernet based network using a minimum of 12 core multimode fibre optic (F/O) cables (except where shown otherwise on Drawings) between each node, capable of operating at 1 Gb/s;
 - (ii) Have a ring configuration from Process XLV Room to the field and back to the Process XLV Room. Some field networks may include multiple rings;
 - (iii) Include field network cables for interface to the PLC panels provided by this Contract; and

- (iv) Routes for cables forming each side of a field network ring, shall be geographically diverse where possible. Where a geographically diverse cable route is not possible then there shall be a minimum of 0.5m separation between the cables which shall be supported and enclosed separately.

P6.7.3 Connections to PAN Subring Network

- (a) The Contractor shall provide a minimum of two 6 core multi-mode fibre optic cables connecting each THP Train unit PLC to the redundant PAN sub-ring network .
- (b) The Contractor shall provide a minimum of one 6 core multi-mode fibre optic cable connecting each compressed air system controller(s) to the redundant PAN sub-ring network.
- (c) The Contractor shall provide a minimum of two 6 core multi-mode fibre optic cables connecting each Sidestream Nitrogen Removal system master PLC and each Sidestream Nitrogen Removal aeration blower master control panel to the respective PAN backbone switches.
- (d) The Contractor shall provide a minimum of one 6 core multi-mode fibre optic cable connecting each Sidestream Nitrogen aeration blower unit LCP to the respective PAN backbone switches.
- (e) Sidestream Nitrogen Removal master PLCs and Sidestream Nitrogen Removal aeration blower master control panel (MCP) shall have two network connections to facilitate connection to the redundant PAN network. For Sidestream Nitrogen Removal aeration blower unit LCP's, both connections may run within a single cable. Refer also to General Specifications G4 Mechanical.

P6.8 INSTRUMENT AND CONTROL PANELS

P6.8.1 General

- (a) Prior to installation, the Contractor shall seek the S.O.'s approval of the proposed location of each instrument panel, analyser panel, Local Control Station (LCS), RIO Panel, Local Control Panel (LCP), MCP, PLC panels, ECCP panels and any other panel.
- (b) ICA panels shall be constructed using materials and according to requirements as specified in General Specification G6.4 Instrument and Control Panel Construction. The Contractor shall propose coating method for the panels (indoor and outdoor) subject to the S.O.'s approval. The panel colours shall be as follows:
 - (i) Process Systems including PLCs and RIO Panels: Light Grey (RAL 7035);
 - (ii) Non-Process Systems (excluding fire systems) : Black (RAL9011); and
 - (iii) Fire Systems : Red (RAL3000).

- (c) If the Contractor requires any additional panels, for the correct functioning of the Works, the Contractor shall provide and is deemed to include in the Contract Sum.
- (d) Each RIO and LCP panel shall contain a section for terminals and ancillaries dedicated to signals forming part of the field wiring to instruments and actuators.
- (e) Local control stations (LCSs) shall be provided for valves, slidegates, and penstocks as specified in the P&IDs. For inaccessible actuated valves (e.g. above 1500mm), inaccessible slidegates, and inaccessible penstocks, LCSs shall be of the remote type and shall be placed in accessible locations within operators' reach. As a minimum, the LCS shall have open and close control buttons, indicating LED lights, local/remote switches for control as specified in the P&IDs. Refer to Particular Specifications P5 – Electrical for Local Control Stations for motors.
- (f) All signals from field devices, instruments, analysers and field equipment shall be connected to I/O modules in designated panels as per the performance requirements specified in this Contract and shall satisfy the Contractor's Design and Build obligations.
- (g) A heat dissipation calculation shall be provided by the Contractor for each RIO, ECCPs, MCPs and LCP panel and submitted for S.O.'s approval.
- (h) Panel-mounted FOBOT's with required FOBOT panels shall be provided for all F/O cables to and/or from all ICA panels provided in this Contract. These FOBOT's are not necessarily shown in the Drawings.

P6.8.2 Outdoor ICA Panels

- (a) The minimum requirements for outdoor ICA panels shall comply with General Specifications G6.4 Instrument and Control Panel Construction.
- (b) PLC control panels (i.e. for THP Train and aeration blower) installed at outdoor locations shall be provided with ventilation to allow for sufficient heat dissipation.
- (c) On-site HMIs on outdoor control panels shall be installed with sufficient protection against direct sunlight.
- (d) ICA panels installed outdoors shall have a sunshade. Sunshades shall be separated from the top of the panel and shall be inclined upwards towards the front and shall overhang the complete perimeter of the equipment by at least 200 mm. Sunshades shall be 316 stainless steel not less than 2 mm thick.
- (e) Outdoor panels shall be installed on a concrete slab and plinth.
- (f) Outdoor panels shall include all panels located in rooms with permanent roofs but without enclosures on all four (4) sides.

P6.8.3 RIO Panels

- (a) RIO panels shall be provided as per the Contractor's Design and Build obligations.
- (b) Power supply for interrogation, interposing relays and solenoids shall be provided from redundant power supply modules located in PLC Panels and RIO Panels.
- (c) Separate wiring and circuit breakers shall be provided for each power supply input connection of any network hardware, located within RIO Panels.
- (d) Panel-mounted FOBOT's shall be provided inside RIO panels for all F/O cables to and/or from the RIO panel. These FOBOT's are not necessarily shown in the Drawings.
- (e) The Contractor shall finalise the I/O requirements of their proposed equipment and design RIO panels accordingly.

P6.9 INSTRUMENTATION

- (a) The Contractor shall provide all instrumentation necessary for the correct, efficient and safe operation of the plant as per the approved Piping and Instrumentation diagrams (P&ID) and as per the Contractor's Design and Build obligations.
- (b) Uniformity of make shall be followed throughout for each type of instrument . E.g. all pH analysers shall be of the same make.
- (c) Prior to installation, the Contractor shall seek the S.O.'s approval of the proposed location of each instrument.
- (d) Power supply for transmitters, contact interrogation, interposing relays and solenoids shall be 24V DC.
- (e) Electronics located in the field shall be conformal coated and enclosed in dust and weather proof cabinets suitable for the environment.
- (f) Instrument installation shall ensure an easy access and retrieving of field instruments for maintenance.
- (g) The Contractor shall ensure instrument installation comply with instrument accuracy as specified in the Specifications. Any additional accessories shall be provided in instrument installation to meet the instrument accuracy as required for functioning of complete system.
- (h) Thermal mass flowmeter shall be of retractable type with an isolation valve. Flow conditioner shall be provided according to manufacturer's recommendation where straight pipe length requirement is not achieved. Proposed flow conditioner shall be submitted and approved by the S.O..

- (i) Five (5) point calibration for electromagnetic flowmeters shall be based on calibrated (measurement) range as per instrument factory standard recommendation.
- (j) Pipe reducer shall be provided according to manufacturer's recommendation for inline instrument where straight pipe length requirement is not achieved. Proposed pipe reducer shall be submitted and approved by the S.O..
- (k) For level measurements in processes that cause thick foam greater than 0.5 meter, the Contractor shall provide radar level transmitter employing 80GHz frequency or higher.
- (l) All flange mounted pressure and differential pressure transmitters utilised for pressure or level measurement shall be provided with a flushing ring.
- (m) For analytical instrument without HART function, a fault contact shall be provided from the analytical instrument connected to the SCADA system via the appropriate PLC, subject to the S.O.'s approval.

P6.9.2 Analyser Panels

- (a) The Contractor shall provide analyser panels as shown on Drawings for mounting of analysers and their sampling accessories. Analyser panels having more than one analyser shall have sufficient spare space and connection points to add one additional analyser.
- (b) Online analysers shall be provided with the necessary sample conditioning system, as recommended by the manufacturer, to minimise sensor maintenance and achieve the manufacturer's specified performance.
- (c) The Contractor shall provide a sample conditioning system (SCS) which may include elements such as filters, debubblers, coolers, as required for analyser protection and sampling line chokage prevention. Valving for sample flow and pressure adjustment shall be as specified in General Specifications G6.12 Instrumentation.
- (d) Sample systems shall also comply with General Specifications G4.7 – Sampling Systems.
- (e) The SCS must not change the composition of the sample, other than to remove unwanted impurities. The SCS shall be implemented either on the common sample line, and/or on lines to individual analysers, as required.
- (f) The arrangement of the sample lines and SCS shall take into consideration the sample lag time and the need for routine maintenance of the SCS.

P6.9.3 Instrument Power Supply

- (a) Instruments that are not loop-powered shall be either:
 - (i) Cabled to a suitable Field UPS DB where there shall be MCB and fuse terminal dedicated to each instrument; or
 - (ii) Connected to a dedicated 24 Vdc fuse terminal in the relevant remote I/O panel.

P6.9.4 Vendor Package Design and Build Instrumentation

- (a) The Contractor shall provide instrumentation for vendor package design and build facilities in this Contract for correctly functioning and fully automated of complete system.
- (b) Instrumentation provided shall comply with G6.12 Instrumentation and P6.9.

P6.9.5 Independent Rooms Temperature Sensors

- (a) Room temperature sensors shall be independent of the BMS and HVAC control systems, to allow independent monitoring and alarming in SCADA as protection against any undetected building services failure.
- (b) Sensors shall be connected using 4-20mA signals.
- (c) Sensors shall be mounted at approximately 1800mm above floor level, and located clear of point sources of heat or cooling, and away from doors, so as to obtain a representative measurement of the room temperature.
- (d) The room temperature sensors shall be connected to RIO panels. Sensors shall be connected using 4-20mA signals.
- (e) Room temperature sensors shall be provided in following rooms:

FACILITY	ROOM NAME:	QUANTITY
WST02	Process XLV Room	1
	Non-Process XLV Room	1
	UPS Room	1
	Battery Room	1
	LV Switchroom 1	1
	LV Switchroom 2	1

P6.10 INFRASTRUCTURE FOR WIRELESS ACCESS POINT NETWORK

- (a) The infrastructure for the Wireless Access Point Network will also serve for the Personnel Tracking System and for future smart facilities systems.

P6.10.1 Infrastructure Locations

- (a) The Contractor shall provide infrastructure for use by Wi-Fi access points and personnel tracking system, at the following locations:

Facility	Indoor Location	QTY	Outdoor Location	QTY
WSB01 Biosolids Building	THP Feed Pump Room 1	1	THP Courtyard	4
WST02 Sidestream Nitrogen Removal	Pump Gallery	2	N removal building Perimeter	4
	LV Switchroom 1	1	Process Area	4
	LV Switchroom 2	1		
	Process XLV Room	1		
	Non-Process XLV Room	1		
	UPS Room	1		
	Battery Room	1		
	Operations Room	1		
	L2 Corridor	1		
C4B Contract Total		11		12

P6.10.2 Infrastructure Requirements

The following infrastructure shall be provided at each of the locations listed above:

- (a) A Wireless Access Point (WAP) Panel (Field Fibre Optic Termination Panel - FFOTP) suitable for application environment with minimum IP56 rating. This panel shall also serve as infrastructure for the personnel tracking system and for future smart facilities. For outdoor WAP antenna locations, the WAP Panel (FFOTP) shall be located at the nearest accessible indoor location;
- (b) A 12 core multi-mode F/O cable, from a FOBOT in Non-Process XLV Room to each WAP Panel;
- (c) A 6A single phase UPS power circuit from field UPS DB to socket outlet within WAP field panel (for future equipment);
- (d) Wi-Fi network hardware equipment and access points (WAP) will be provided by C5A; and
- (e) Personnel tracking system hardware will be provided by C6B.

P6.11 POWER MONITORING SYSTEM (PMS)

- (a) This Contractor shall provide the following PMS components:
 - (i) Energy meters for process and non-process switchboards as specified in Particular Specification P5 Electrical Works and General Specifications G5 Electrical;
 - (ii) Edge network switches for PMS at process and non-process switchboards shall be housed in wall-mounted switch rack and shall comply with General Specifications G6.20 Data Communication Networks and have their uplink connected to a port on a Non-Process Backbone switch. Non-Process Backbone switches are by others (C5A);
 - (iii) Ethernet communications and connections between each energy meter, protection relay and switchboard temperature monitoring sensor to PMS edge switch as shown in the typical PMS schematics; and
 - (iv) PMS I/O modules located within process and non-process switchboards and MCC's to collect status signals from MCCB's.
- (b) Data from energy meters, protection relays, switchboard thermal monitoring system, battery monitoring system, generator, ATS and PMS I/O modules in this Contract will be interfaced with a sitewide PMS by C5A, via the Non-Process Backbone.
- (c) The Contractor shall provide PMS data interfaces at intelligent motor control centres which are compatible with the PMS system being provided by C5A. PMS system shall use Modbus protocol.
- (d) This table describes the signal types and collection methods which shall be provided by the PMS. The Contractor shall include any other signals as specified in Particular Specification P5 Electrical Works and General Specification G5 Electrical for monitoring at PMS.

TUAS WATER RECLAMATION PLANT
CONTRACT 4B - DIGESTION PRE-TREATMENT AND
VOLUME 3 SIDESTREAM NITROGEN REMOVAL FACILITIES (2ND-STAGE)

Panel:	Breaker Type:	Signal From	Interface with SCADA / PLC:				Interface with PMS:		
			SCADA Control	SCADA Monitoring	Signal type to SCADA	Signal Source	PMS Monitoring	Signals Type to PMS	Signal Source
400V SWGR and MCC (800 A and above)	ACB	Incoming feeder Outgoing Feeder Tie-breaker	N/A	N/A	N/A	N/A	Opened Closed Trip	Ethernet	Protection relay
		Energy Meter	N/A	N/A	N/A	N/A	Energy Meter	Ethernet	Energy Meter
		Temperature Monitoring for switchboard/ switchgear/ MCC	N/A	N/A	N/A	N/A	Temperature Monitoring for switchboard / switchgear/ MCC	Ethernet	Temperature Monitoring for switchboard/ switchgear/ MCC
400V SWGR, MCC & MDB (Below 800 A)	MCCB	Incoming feeder Outgoing Feeder Tie-breaker	N/A	N/A	N/A	N/A	Opened Closed Trip	Hardwired to PMS I/O	MCCB
		Energy Meter	N/A	N/A	N/A	N/A	Energy Meter	Ethernet	Energy Meter
		Temperature Monitoring for switchboard/ switchgear/ MCC	N/A	N/A	N/A	N/A	Temperature Monitoring for switchboard / switchgear/ MCC	Ethernet	Temperature Monitoring for switchboard/ switchgear/ MCC
Lighting Distribution Board	N.A	Energy Meter	N/A	N/A	N/A	N/A	Energy Meter	Ethernet	Energy Meter

P6.12 ASSET MANAGEMENT SYSTEM (AMS)

- (a) Refer to General Specifications G1 General for detailed requirements.
- (b) Contractor shall submit documentation of all installed assets and liaise with the Board and S.O. for computerised asset listing and data integration with the Board's AMS according to requirements in General Specifications G1 General.
- (c) Contractor shall comply with Particular Specification P13 Asset Management and Maintenance System to ensure interoperability with the Board's AMS.

P6.13 TESTING & COMMISSIONING

- (a) Panels shall be tested in accordance with General Specifications G6.4 Instrument and Control Panel Construction.
- (b) Instrumentation shall be factory tested and calibrated in accordance with General Specification G6.12 Instrumentation. Analytical instruments shall be field-calibrated during commissioning, and at regular intervals thereafter until Practical Completion.
- (c) Fibre optic cables shall be tested in accordance with IEC61280 as follows:
 - (i) Factory Test Results shall be furnished, showing attenuation at two operating wavelengths for each fibre in each cable reel. These results shall be indexed to identify where each reel was installed in the network;
 - (ii) After installation and termination, bi-directional testing of every fibre shall be completed using an Optical Time Domain Reflectometer (OTDR) at 850 nm for multimode fibre or 1310 nm and 1550 nm for single mode fibre. The test results shall be supplied to the S.O. in hard and soft copies. Any software required to view the OTDR traces shall also be supplied. It is expected that the test results will match the factory tests for the cable reels that have been installed; and
 - (iii) In the event that the test results indicate point discontinuities at locations other than terminations, or that the installed fibre links do not meet the attenuation requirements defined herein, it shall be the Contractor's responsibility to correct the problem.
- (d) Copper data cables shall be tested in accordance with Telecommunications Industry Association (TIA) standards.
- (e) Power and signal cabling shall be tested in accordance with General Specifications G5 Electrical.
- (f) Signal testing from all field devices provided by this Contract all the way to /from SCADA shall be carried out in cooperation with C4A and C5A.
- (g) Testing and commissioning shall be carried out jointly with C4A/C5A. The Contractor shall provide personnel for the testing and commissioning of all the equipment installed by the Contractor, including personnel allocated to work with the S.O.'s control MCS programmers to verify the correct sequencing and operation of the Contractor's equipment.
- (h) All testing and commissioning shall also be in accordance with General Specifications G2 Preliminaries.

P6.14 CONTRACTORS SUBMITTALS

- (a) The Contractor shall submit the following documentation covering equipment where final design is by the Contractor:
- (i) Submittals required to document the Contractor's vendor package design and build final System Architecture design, refer to P6.3, P6.4 and P6.6;
 - (ii) Submittals required for Digestion Pre-treatment Facility and Sidestream Nitrogen Removal Facility SCADA on-site HMIs' graphical presentation and operations, including operator selections, operating parameter settings, alarms, trends, etc;
 - (iii) PLC I/O Schedule showing rack, slot, channel, spare channels, spare slots. Summary sheet shall show number of spares in each panel by I/O type and a bill of materials;
 - (iv) Panel internal power distribution schematics;
 - (v) Panel internal and external GA drawings;
 - (vi) I/O module wiring diagrams;
 - (vii) Intrinsic safety cable calculations;
 - (viii) Instrument mounting drawings;
 - (ix) Safety Instrumented Functions (SIF level) design calculations;
 - (x) Fibre optic power loss calculations for field control networks;
 - (xi) Fibre-core patch register;
 - (xii) IP address list;
 - (xiii) Cable routing and all instrument location details in BIM including cable routing and field instrumentation within package unit vendor package design and build system;
 - (xiv) Detailed instrument register including vendor package design and build field instrumentation in BIM with makes and models; and
 - (xv) Functional Design Specification (FDS) for any vendor package design and build PLCs; and
 - (xvi) Any submittals required to satisfy Cyber Security Requirements.
- (b) Refer to Particular Specifications P2 Preliminaries and General for additional submittal requirements.

P6.15 APPENDICES

- (a) APPENDIX P6A – Digestion Pre-treatment and Sidestream Nitrogen Removal PCN.
- (b) APPENDIX P6B – Standard Control Features PCN.

END OF SECTION